

instruments are being used for state-of-the-art electronic systems such as RADAR, satellite and surveillance. Shortening the product development cycle has long been a key objective of military and aerospace organisations. One method to reduce the development time is concurrent design and test, which is often represented with the V-diagram product development model. In these industries, for which the end product is a highly complex 'system of systems,' the left side of the V-diagram is considered 'design' and the right side represents 'test.'

Defence communication systems have traditionally been analogue in nature. However, digital technologies are replacing the traditional analogue communication systems. To successfully implement the digital systems, one needs to analyse the technology manifestations in different environments. Software-defined and modular test systems can be used to meet these requirements.

Catering to each technology, there may be various testing needs. For instance, in communication one may have to check the signal frequency, power, harmonics, etc, while in automotive one may have to do different levels of stress testing, torque testing, pressure testing, etc.

As wireless communication continues to grow, the pressure to find innovative ways to more effectively use channel bandwidth is increasing. One of the most recent innovations is the development of multiple-input multiple-output (MIMO) antennae systems to improve the signal-to-noise ratio and increase the throughput. Accurate testing of MIMO transceivers presents significant challenges to existing test instrumentation architectures. New architectures like hybrid PXI or benchtop instrumentation enable not only advanced signal processing algorithms to multiplex and de-multiplex various spatial streams but also tight synchronisation between transmit and receive antennae.

In general, T&M users are increasingly demanding the best possible measurement and source accuracy,



N9344C handheld spectrum analyser by Agilent Technologies

noise, signal response, and system throughput for today's demanding measurements. With the constant evolution of technology, they expect upgradeable instruments with backward technology support. Also, as data rates increase, there is demand for higher-bandwidth instruments. In the manufacturing domain, fully automated environment test setups are in demand. Similarly, as the communication technologies move from 2G to 3G to 4G, one needs handheld, portable but rugged instruments to deploy base stations and towers.

Users want the T&M instruments to be capable of offering universal connectivity via GPIB, USB, LXI/Ethernet and other interfaces. They increasingly expect T&M products to keep pace with rapid evolutions in desktop, mobile and Internet computing standards, the most recent of which is support for mobile devices like smartphones and tablets. The software that enables all of these advanced capabilities must be simple and easy to understand, use and integrate into human interactive and programmatic applications in the lab and the factory.

Different instruments at a glance

Oscilloscopes by and large have seen most changes over the last couple of years.

"With the availability of high-end

technology like digital signal processors and field-programmable gate arrays at low cost, signal acquisition has become pretty fast. This has made it possible to get real-time display on the LCD of a digital oscilloscope, unlike in cathode-ray tube oscilloscopes. With digital oscilloscopes, it is also possible to add advanced features like fast Fourier transform, waveform recorder, data logger, advance triggering and logic channels at a very low cost," says P. Prabhu, general manager (technical), Scientific Mes-Technik.

Other test instruments like digital multimeters, function generators and frequency counters have also shrunk in size with advanced features. Microcontroller-based designs in these instruments have helped in reducing the number of mechanical switches, thereby improving the product's performance and life.

With automation for mass and quality product manufacturing, there is increase in the use of measuring instruments in production lines. Sometimes the measurement data needs to be collected and analysed in less than 10 seconds. Multiple data needs to be collected while the product is on the conveyor belt for a very short duration. These challenges are being met with new instruments.

Specifically, general-purpose T&M instruments have witnessed the following changes in terms of technology: